

RETAIL SALES ANALYSIS

1. Project Overview

Analyze sales performance, customer behavior, payment trends, logistics performance, and customer satisfaction using Excel, Power Query, Pivot Tables, and interactive dashboards.

2. Dataset Overview

Dataset	Business Purpose	Primary Key	Foreign Key	Can Duplicate Rows Exist?	One Row Represents
Customers	Stores customer location information	customer_id	None	No (based on the customer_id)	One customer record
Orders	Stores information about every customer order	order_id	customer_id	no (based on order_id)	One order
Order Items	Stores the products included in each order	order_id + order_item_id (Composite Primary Key)	order_id, product_id, seller_id	Yes, an order_id can repeat because one order can have multiple items	One product within one order
Products	Stores information about each product	product_id	none	no (based on product_id)	One product within one order
Sellers	Stores information about each seller	seller_id	None	No (based on seller_id)	One seller
Payments	Stores payment details for each order	No single primary key (can be treated as order_id + payment_sequential)	order_id	Yes, one order can have multiple payments	One payment transaction for an order
Reviews	Stores customer reviews and ratings	review_id	order_id	review_id: No; order_id: Yes (some orders may have multiple reviews in this dataset)	One review record
Geolocation	Stores latitude and longitude for zip codes	No reliable primary key	None	Yes, the same zip code can appear multiple times with different coordinates	One geolocation record
Category Translation	Translates Portuguese product category names into English	product_category_name	None	No	One category translation

3.Entity Relationship Overview

Parent Table	Parent Key	Child Table	Child Key	Relationship
Customers	customer_id	Orders	customer_id	One-to-Many
Orders	order_id	Order Items	order_id	One-to-Many
Orders	order_id	Payments	order_id	One-to-Many
Orders	order_id	Reviews	order_id	One-to-Many
Products	product_id	Order Items	product_id	One-to-Many
Sellers	seller_id	Order Items	seller_id	One-to-Many
Geolocation	zip_code_prefix	Customers	customer_zip_code_prefix	One-to-Many
Geolocation	zip_code_prefix	Sellers	seller_zip_code_prefix	One-to-Many
Category Translation	product_category_name	Products	product_category_name	One-to-Many

4. Initial Data Quality Assessment

Dataset	Primary Key	Rows	Columns	Duplicate PK	Missing Values
Customers	customer_id	99,441	5	No	No
Geolocation	No Primary Key	10,00,163	5	Yes	No
Order Items	order_id + order_item_id	1,12,650	7	No	No
Payments	order_id + payment_sequential	1,03,886	5	No	No
Reviews	review_id	99,224	7	No	Yes
Orders	order_id	99,441	8	No	No
Products	product_id	32,951	9	No	Yes
Sellers	seller_id	3,095	4	No	No
Category Translation	product_category_name	71	2	No	No

5. Merge Process

Power Query ETL Workflow	Parent Query	Merged Table	Join Key	Columns Added	Business Purpose
1	Order Items	Orders	order_id	customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date	Add customer reference, order status, purchase date and delivery timeline for order analysis.
2	Master_Data	Customers	customer_id	customer_city, customer_state	Analyze sales by city/state and prepare for customer geolocation merge.
3	Master_Data	Products	product_id	product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm	Product category analysis, freight analysis, product dimension analysis.
4	Master_Data	Category Translation	product_category_name	product_category_name_english	Convert Portuguese category names into English for dashboards and reports.
5	Master_Data	Aggregated Payments	order_id	payment_installments, payment_value	Revenue and installment analysis while preventing duplicate rows caused by multiple payment records.

6	Master_Data	Original Payments Table	order_id	payment_type (used only for Payment Method analysis)	Preserve payment method information for the Payment Method visualization.
8	Master_Data	Aggregated Reviews	order_id	review_score	Customer satisfaction analysis while preventing duplicate review records.
9	Master_Data	Sellers	seller_id	seller_city, seller_state	Seller location analysis.

6. Data Cleaning & Transformation

Step	Action	Reason
1	Fixed oversized Excel table ranges	Prevented Power Query from importing blank rows
2	Corrected data types	Ensured proper joins and calculations
3	Aggregated Payments by order_id	Prevented one-to-many row duplication
4	Aggregated Reviews by order_id	Prevented duplicate review records
5	Excluded Geolocation from Master_Data	Avoided row multiplication due to one-to-many ZIP relationships (Geolocation was intentionally excluded from the Master_Data table because ZIP codes have a one-to-many relationship with coordinates, which could multiply rows. The table was retained separately for future geographic visualization or Power BI mapping.)
6	Ignored blank product categories in KPI	Ensured accurate category count

7. KPI Definitions

KPI	Formula / Definition
Total Revenue	Sum(payment_value)
Total Orders	Distinct Count(order_id)
Total Customers	Distinct Count(customer_id)
Average Order Value	Revenue / Orders
Average Review	Average(review_score)
Average Delivery Days	Delivered Date – Purchase Date
Total Sellers	Distinct Count(seller_id)
Products Sold	Count(order_item_id)

8. Dashboard Overview

Chart	Business Question Answered
Revenue Trend Over Time	How has revenue changed over time?
Revenue by Product Category	Which categories generate the highest revenue?
Revenue by Customer State	Which states contribute the most revenue?
Order Status Distribution	What percentage of orders are delivered, canceled, or processing?
Revenue by Payment Method	Which payment methods generate the highest revenue?
Payment Installment Distribution	How do customers prefer to pay?
Review Score Distribution	What is the customer satisfaction level?
Average Delivery Time by Customer State	Which states experience longer delivery times?

9. Business Questions & Findings

Business Question	Answered By	Key Finding
How has revenue changed over time?	Revenue Trend Over Time	Revenue showed consistent growth before stabilizing.
Which product categories generate the highest revenue?	Revenue by Product Category	Bed Bath Table and Health Beauty were among the top revenue-generating categories.
Which customer states contribute the highest revenue?	Revenue by Customer State	São Paulo generated the highest revenue.
What is the order fulfillment status?	Order Status Distribution	Most orders were successfully delivered.
Which payment methods generate the most revenue?	Revenue by Payment Method	Credit Card contributed the highest revenue.
How do customers prefer to pay?	Payment Installment Distribution	Most customers preferred paying in a single installment.
How satisfied are customers?	Review Score Distribution	Most reviews were rated 4 or 5 stars.
Which states experience longer delivery times?	Average Delivery Time by Customer State	Delivery time varied significantly across states.

10. Tools Used

Tool / Feature	Purpose	Skills Demonstrated
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Microsoft Excel 2021	Data analysis, dashboard development, and reporting	Excel Analytics, Spreadsheet Design
Power Query	Data extraction, cleaning, transformation, and merging	ETL, Data Cleaning, Data Transformation
Pivot Tables	Data summarization and KPI calculations	Data Aggregation, Business Analysis
Pivot Charts	Interactive data visualization	Data Visualization, Dashboard Design
Slicers	Interactive filtering of dashboard reports	Interactive Reporting, User Experience
Excel Tables	Structured data management	Data Organization, Structured References
Power Query Merge & Append	Combined multiple related datasets	Data Integration, Relational Data Handling
Power Query Group By	Aggregated Payments and Reviews to avoid duplicate records	Data Modeling, Data Preparation
Calculated Columns & Excel Formulas	KPI calculations and delivery time analysis	KPI Development, Analytical Thinking
Documentation	Recorded ETL workflow, relationships, and business logic	Technical Documentation, Business Intelligence

Core Skills Gained : ETL • Data Cleaning • Data Transformation • Data Modeling • Business Intelligence • Dashboard Design • KPI Development • Data Visualization • Interactive Reporting • Technical Documentation

Project Outcome

Successfully developed an interactive Retail Sales Analysis Dashboard by integrating multiple relational datasets using Power Query, performing ETL and data cleaning, creating business KPIs, and designing an interactive dashboard with PivotTables, PivotCharts, and Slicers. The project demonstrates end-to-end Excel Business Intelligence skills, from raw data preparation to executive-level reporting.